



Drivers of atmospheric CO₂ concentration in southeast Brazil: Insights from land use change, vegetation, and climate factors

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ABSTRACT

The atmospheric carbon dioxide (CO₂) increase is one of the main causes of climate change and has been a global concern. Due to the world's necessity of understanding CO₂ sources and sinks, satellites such as the Orbiting Carbon Observatory 2 (OCO-2) have been launched to monitor greenhouse gases (GHG). This study aimed to characterize the temporal variability and the main factors controlling (i.e., vegetation, climate, and land use changes) the column-averaged atmospheric CO₂ (Xco₂) in Southeast Brazil (925 thousand km²) between 2015 and 2021. Xco₂ was detrended by subtracting the world's trend to study regional features (Xco_{2R}). The vegetation component was sampled through the Solar Induced Chlorophyll Fluorescence (SIF) and Gross Primary Production (GPP). As regards climate parameters, we used Precipitation (Prec) and Global Irradiation (Qg), while the Land Use Change (LUC) was characterized by using the land use classification maps available to the region. Xco_{2R} follows a seasonal pattern of maximum averages during autumn-winter and minimum averages spring-summer, with the highest value observed in October 2021 (393.4 ± 1.26 ppm) and the lowest in April 2015 (389.37 ± 1.7 ppm). For the other variables, this seasonality is inverted which means those have negative linear correlations with Xco_{2R} ranging from −0.35 to −0.6. We observed an Xco_{2R} temporal increase trend since 2017 and this tendency is higher over forest areas which could be related to deforestation observed in the same period. In addition, the emission data from the national GHG inventory estimated an increase of CO₂ emissions due to land use changes in the same period and was able to explain around 50 % of the annual Xco_{2R} increase in Southeast Brazil. Thus, vegetation has an essential role in the Xco_{2R} absorption, and especially the land use change and deforestation were the main CO₂ sources to the atmosphere on a regional scale.

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1. Introduction

Climate change is a global concern and one of the main causes is the increase of greenhouse gases (GHG), especially carbon dioxide (CO_2). The CO_2 reached 423 ppm of global atmospheric concentration in 2024 according to the National Oceanic and Atmospheric Administration (NOAA and AuthorAnonymous, 2024). This increase is mainly related to fossil fuel combustion (Hedelius et al., 2018), but in Brazil Land Use Change (LUC) has been the main cause of CO_2 emission (De Azevedo et al., 2018; Garofalo et al., 2022; C. H. L. Silva et al., 2020). Moreover, LUC in Brazil is historically related to farming expansion, especially deforestation and pasture opening (Souza et al., 2020). Recently, there have been records of deforestation all over the Brazil territory (Escobar, 2019; Silva et al., 2020).

In the Southeast region, the Land Use (LU) is mostly composed of farming areas, namely pasture. Meanwhile, the forest formations generally are small and scattered forest fragments or savannas (de Mendonça et al., 2022; Parras et al., 2020; Souza et al., 2020). This composition of LU gives the Southeast an overall vegetation seasonal cycle dependent on farming activity. Studies conducted in some areas inside the Southeast region have been reporting this condition (da Costa et al., 2022a; Morais Filho et al., 2021). For example, da Costa et al. (2022a) studied the spatiotemporal behavior of Xco_2 with satellite data in a sugarcane-producing region and Morais Filho et al. (2021) studied the temporal behavior of the Xco_2 also with satellite data in three different types of crops, including sugarcane. They argued that the atmospheric CO_2 concentration generally has higher concentrations during the harvest and lowest during growing seasons in producing areas. In spite of showing an emission profile quite related to land use and land use change, fire emission also is pivotal for the Brazil (Gomes et al., 2024). Additionally, densely populated also contribute with fossil fuel burning and industry (Hamilton et al., 2024; Labzovskii et al., 2019).

Among the main sinks of atmospheric CO_2 , vegetation cover is paramount through photosynthesis (Chhabra and Gohel, 2019; Li et al., 2020). However, since this process is water and light-dependent, the climate conditions could shape this relationship (Barbosa et al., 2015; da Costa et al., 2023; Golkar et al., 2019). Odorizzi de Campos et al. (2022), aiming to understand the impact of managed and degraded pasture in a Cerrado domain, observed that Xco_2 between 2015 and 2020 has a direct relation with land surface temperature amplitude and the sensitivity to this amplitude is higher in a degraded pasture, thus being less resilient to future climate changes.

Given the global necessity to understand CO_2 sources and sinks (IPCC, 2022, 2021), satellite missions have been developed to monitor this and other GHG (Crisp et al., 2017; Eldering et al., 2019; Moore et al., 2018; Yokota et al., 2009). Among these missions, the Orbiting Carbon Observatory – 2 (OCO-2) was launched in 2014 to provide observations regarding the column-averaged atmospheric CO_2 (Xco_2 , Crisp et al., 2017). And, due to the instruments design, this satellite is also capable of retrieving the Solar Induced Chlorophyll Fluorescence (SIF, Frankenberg et al., 2014) that has been used as a proxy of photosynthesis (Mohammed et al., 2019; Sun et al., 2018). In this sense, global studies seek to identify possible sources and sinks of CO_2 in the atmosphere (Hakkarainen et al., 2016, 2019). Recently, Santos et al. (2022), using Xco_2 anomalies, a measure of possible sources and sinks, studied the spatial variability of these anomalies in the Maranhão's eastern Amazon. They found significant hotspots (possible sources) of CO_2 in the southeast of the studied region between 2015 and 2018, but they did not investigate the role of land use changes. On the other hand, Crivelari-Costa et al. (2023) reported a reduction of around 5 Tg C in carbon absorption due to deforestation in the Brazilian Legal Amazon between 2009 and 2019. Nevertheless, remote sensing-based emission estimates are not trivial, as CO_2 vertical's profile and atmospheric transport introduce uncertainties in those estimates (Hamilton et al., 2024; Wang et al., 2024).

Despite all the spatial and temporal coverage provided by satellite imagery, few studies have been conducted to investigate the role

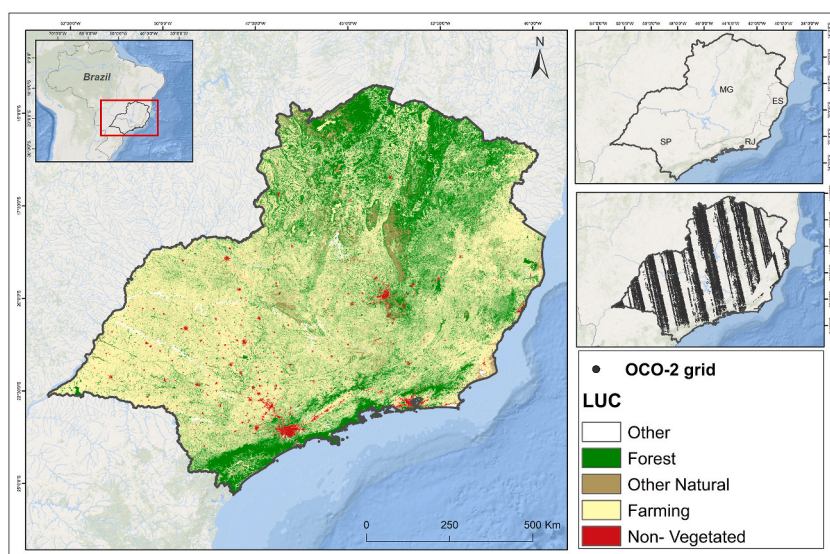


Fig. 1. Study area overlaid by the land use occupation in 2021 (source: MapBiomass, 2023); SP – São Paulo, ES – Espírito Santo; RJ – Rio de Janeiro, MG – Minas Gerais.

of land use on the dynamics of CO₂ concentration at a regional scale, especially in the tropics. In this sense, this study aimed to characterize the temporal variability and the main factors controlling the column-averaged atmospheric CO₂ (Xco₂) in Southeast Brazil. Specifically, we evaluate how climate, vegetation and land use change affect the regional Xco₂, disregarding atmospheric transport.

2. Material and methods

2.1. Study area

The work was conducted in the Southeast (SE) region of Brazil (Fig. 1) which covers an area of 925,000 km² composed of four states: São Paulo (SP), Rio de Janeiro (RJ), Espírito Santo (ES), and Minas Gerais (MG). The Brazilian Southeast has the highest population density - over 87 million inhabitants - as well as the largest share in agricultural production outcome (27 %) that represents 55 % of the country's gross domestic product (Ipeadata, 2020).

The SE region intersects the domains of Atlantic Forest (tropical rainforest) and Cerrado (savannah) biomes. The region's climate can be broadly classified into three groups: humid subtropical with dry winter; humid tropical with dry winter; and sub-humid tropical with dry winter (Alvares et al., 2013). Typically, the seasons of spring-summer starts on October/November and ends at March/April, complementary with the fall-winter season that starts on April/May and goes until September/October.

The map of land use and occupation (Fig. 1) shows that 62.8 % of all the SE is composed of farming, which includes agriculture and pastures. Forest cover occupies around 29.4 % and other natural formations around 4.4 %. Non-vegetated areas cover around 1.9 % while the water bodies occupy around 1.5 % (MapBiomass, 2023).

2.2. Remote sensing data acquisition

In this study, we used meteorological, vegetation and land use data from different sources to understand their impact on the column-averaged atmospheric CO₂. The characteristics from each of those products, such as, time and spatial resolution are summarized in Table 1 and later described in this section. The period comprehended in this study started in 2015 and ended in 2021. The reason to stop in 2021 was that the latest version of MapBiomass (v7.1) used to assess land use, as well as the CO₂ emission data from SEEG were not available beyond 2021. It is important to highlight that both database was updated after the completion of this study.

To equilibrate the spatial and temporal resolutions among the orbital sensors used in this study, a flowchart (Fig. 2) established a standard in data acquisition from the other platforms based on the coordinates obtained in the OCO-2 platform. Although the spatial resolution diverges among the datasets, we acquired the data based on OCO-2 observations, meaning that we only used the collocated or the nearest observations from the other platforms with the OCO-2 satellite (da Costa et al., 2022a). Furthermore, after completing the data acquisition all variables were recast to a month scale, mitigating the spatial differences among the sources. Xco₂ pre-process consisted of outliers' removal based on the interquartile rule and detrending (see section 2.3). We emphasize that similar studies using data from satellites with different resolutions have already applied comparable approaches (Golkar et al., 2019; Odorizzi de Campos et al., 2022; Rossi et al., 2022). All pre-processed data used in the analyses, as well as the programming codes, are available at https://github.com/lm-costa/xco2_drivers_2023.

2.2.1. Carbon dioxide data

The column-averaged atmospheric CO₂ concentration (Xco₂) was obtained from the Orbiting Carbon Observatory 2 (OCO-2)

Table 1

Studied variables, sensors, and data platforms, combined with their respective spatial and temporal resolutions.

Variable	Acronym	Unit	Database	Temporal resolution	Spatial resolution
CO₂					
Column averaged of atmospheric CO ₂ CO ₂ equivalent emission	Xco ₂ ECO ₂	ppm t 10 ⁶	OCO-2 SEEG	16 days annual	1.29 km × 2.25 km –
Climate					
Global Solar Irradiation	Qg	MJ m ⁻² d ⁻¹	FLASH Flux Version 3 (A, B, C) NASA/POWER	Daily	111.3 km × 111.3 km
Precipitation	Prec	mm	GEO-5 FP-IT NASA/POWER	Daily	111.3 km × 111.3 km
Vegetation					
Solar-Induced chlorophyll Fluorescence 757 nm	SIF 757 nm	Wm ⁻² sr ⁻¹ μm ⁻¹	OCO-2	16 days	1.29 km × 2.25 km
Gross Primary Production	GPP	kg C m ² 8day ⁻¹	MYD17A2H v006 MODIS-Aqua	8 days	500 m × 500 m
Land Use					
Land Use	LU	–	MapBiomass	annual	30 m × 30 m

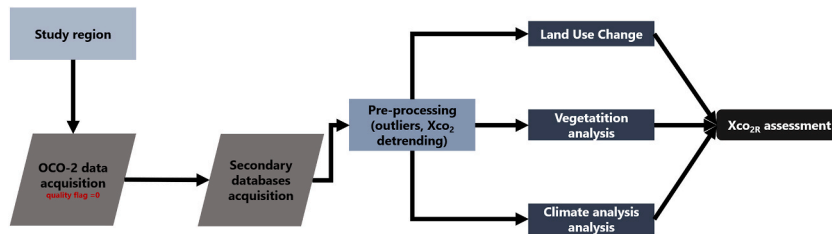


Fig. 2. Flowchart of data acquisition and analysis process.

project. This mission was developed by NASA to remotely monitor the X_{CO_2} on a global scale (Crisp et al., 2012; O'Dell et al., 2012, 2018). The OCO-2 data used here was the level 2 version 10 (https://disc.gsfc.nasa.gov/datasets/OCO2_L2_Lite_FP_10r/summary). It is important to notice that, we only considered the observations without cloud coverage (quality flag = 0) (O'Dell et al., 2018). This satellite has a footprint around 1.29×2.25 km and a temporal resolution of 16 days, where one observation can be composed up to 8 footprints (Crisp et al., 2017; Kataoka et al., 2017).

In this work, we compared remote sensing data with the National greenhouse gas emission inventory (in CO_2 equivalent) published by the Brazilian Gas Emission Estimation System (SEEG, De Azevedo et al., 2018). The SEEG CO_2 equivalent emissions (ECO_2) are estimated based on IPCC methodologies and are divided by Brazil's states and economic sectors, such as farming. In this work, we compared X_{CO_2R} data with those from the Southeast land use change emission sector. This comparison was made to understand if the X_{CO_2} follows the emission pattern related to land use changes. Specifically for this sector, the SEEG uses the MapBiomass classification to estimate emissions related to land use change along Brazil, and it is important to emphasize that SEEG's estimates use the latest version of MapBiomass data, which at the time of this work was the 7.1v. The SEEG data are freely available at: <https://seeg.eco.br/>.

2.2.2. Climatology data

Meteorological variables (Table 1) were collected from NASA POWER daily dataset. The project Predictions of World Energy Resource (POWER, <https://power.larc.nasa.gov/>) initiated by NASA aimed to enhance the current renewable energy dataset and create new datasets from satellite systems (Stackhouse, 2020). The global distribution of precipitation values is obtained from the GPCP 1-DD Satellite-Gauge product and the Integrated Multi-Satellite Retrievals for Global Precipitation Measurements (IMERG, Pradhan et al., 2022), while the solar radiation is from FLASHFlux 4 (García Pedreros and Ospina-Noreña, 2022; Jia et al., 2013; Yan et al., 2011). Although there is other finer resolution meteorological data, NASA-POWER has several products that are systematically available, reducing the bias introduced by different datasets. Moreover, recent studies already showed that NASA-POWER data has a reasonable agreement with the ground stations in Brazil (Carrara et al., 2023; Duarte and Sentelhas, 2020).

2.2.3. Vegetation data

The Solar Induced Chlorophyll Fluorescence (SIF) is a re-emission phenomenon of absorbed light originated in the chlorophyll-a molecules, and this signal has been used as a photosynthesis and vegetation proxy worldwide (Mohammed et al., 2019). Although there is a SIF's product with higher resolution based on MODIS footprint (Sun et al., 2023), we used SIF data from the OCO-2 mission to access the vegetation pattern in Southeast Brazil as this would bring an advantage of being retrieved together with the X_{CO_2} data. Due to the OCO-2 instruments, this satellite is also capable of retrieving the Solar-Induced Chlorophyll Fluorescence at 757 nm and 771 nm (Frankenberg et al., 2014). However, we only evaluated the SIF at 757 nm for two reasons: i) is closer to the far-red peak position (~740 nm; Cogliati et al., 2019; Mohammed et al., 2019); ii) other studies reported that the correlations values with X_{CO_2} were better (da Costa et al., 2022b; Li et al., 2018; Morais Filho et al., 2021). The level 2 data version 10 daily SIF available at 757 nm (https://disc.gsfc.nasa.gov/datasets/OCO2_L2_Lite_SIF_10r/summary) was used in this study.

Besides the SIF, this study evaluated the Gross Primary Production (GPP) as another vegetation proxy. The GPP data was obtained from the Application for Extracting and Exploring Analysis Ready Sample (AppEEARS, <https://appeears.earthdatacloud.nasa.gov/>). The AppEEARS allows users to obtain subsets of large databases using spatial and temporal parameters. Sample requests submitted to the AppEEARS provide users with not only data values but also associated data quality. Interactive visualizations with summary statistics are provided for each sample within the application, which allows users to view and interact with their samples before downloading the data (AppEEARS Team, 2023). Specifically for this study, we requested the Gross Primary Production (GPP) derived from a MODIS sensor (MYD17A2H v006). This product is a cumulative 8 days composite of values based on the radiation use efficiency concept, expressed as: $GPP = \varepsilon \times APAR$

2.2.4. Land use data

To investigate the evolution of land use change, matrix data on land use and cover were acquired from MapBiomass product "Brazilian Annual Land Use and Land Cover Mapping Project", collection 7.1, 30m resolution; spanning the 2015, 2017, and 2021 period. The MapBiomass (<https://mapbiomas.org/>) is a Brazilian initiative launched in 2015 to contribute to the understanding of the land use dynamics, providing annual maps from 1985 up to now (the last version released until our analysis was Collection 7.1). The MapBiomass methodology uses the Landsat image collection and machine learning algorithm to classify the land uses in Brazil with high accuracy, available in an automated interface with Google Earth Engine (MapBiomass, 2023; Souza et al., 2020).

Subsequently, we reclassified the land use into 5 categories: i) Forest: comprehends all the forest formations, including savanna, mangrove, and wooden sandbank; ii) Other Natural: flooded field and swamp (wetlands), natural herbaceous vegetation, and other natural formations; iii) Farming: this land use is composed by pasture, agriculture, silviculture, and other farming uses; iv) Non-vegetated: in this classification was only used the urban and mining areas; v) Others: all other land uses in the Southeast, such as beach, water bodies and unclassified pixels.

Considering that land use in the southeast is mostly composed of farming plus forest mosaics and farming areas, where conversions of natural habitats into anthropic areas are also expressive (Souza et al., 2020), the Land Use Change (LUC) analysis was focused primarily on deforestation due to farming and reforestation over farming areas. The LUC was calculated by the difference between the raster files of 2021, 2017, and 2015. Since more than 90 % of the region is characterized by those land uses (i.e., farming and forest) this approximated well the impact of LUC on X_{CO_2} .

2.3. Temporal analysis

The X_{CO_2} was detrended by a regression method proposed by Gujarati & Porter (2011). This detrend has the purpose to diminish global bias and atmospheric transport or, put another way, to regionalizing the X_{CO_2} ($X_{CO_{2R}}$, Eq. (3)). This was made to study the relationship between the CO_2 concentration and the environmental factors (Albright et al., 2022; Odorizzi de Campos et al., 2022; Rossi et al., 2022). To accomplish this, first the X_{CO_2} was adjusted to a linear regression based on observations (Eq. (1)). Then, the differences between the estimated and observed values were calculated (Eq. (2)) and finally the detrend was made by removing this

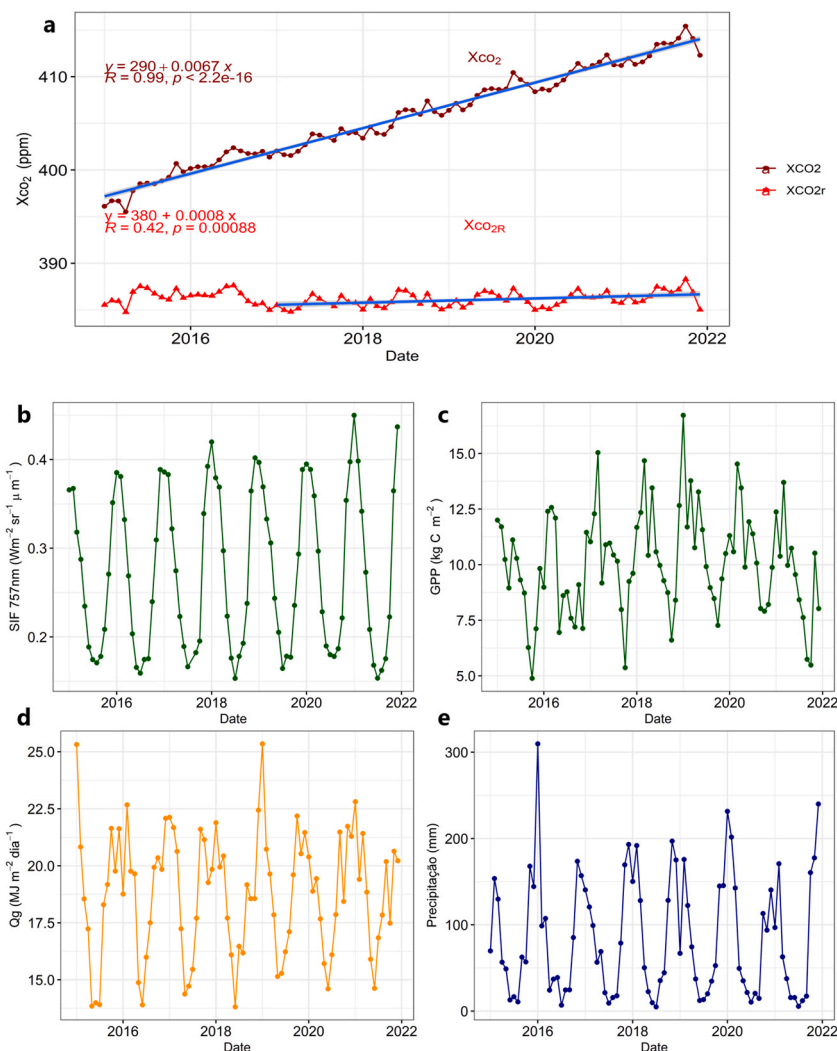


Fig. 3. Temporal variability between 2015 and 2021 of a) X_{CO_2} and $X_{CO_{2R}}$, b) SIF 757 nm, c) GPP, d) Q_g , and e) Prec. Where X_{CO_2} = column-averaged atmospheric CO_2 , $X_{CO_{2R}}$ = regionalized column-averaged atmospheric CO_2 , SIF 757 nm = Solar-Induced chlorophyll Fluorescence 757 nm, GPP = gross primary production, Prec = precipitation, Q_g = global irradiation. Residuals related with $X_{CO_{2R}}$ are presented in SF1a.

difference (Xco_{2R} , Eq. (3)). Although atmospheric transport influence can be reduced though the regionalization process (Albright et al., 2022; da Costa et al., 2022b; Hakkarainen et al., 2016), this process still has influence on Xco_2 , being an uncertainty source for our study.

$$Xco_{2est_i} = \alpha + \beta X_i \quad (\text{Eq. 1})$$

where α = linear coefficient; β = angular coefficient and X_i = i -th observation number.

$$\Delta_i = Xco_{2est_i} - Xco_{2i} \quad (\text{Eq. 2})$$

where Δ_i = difference between the estimated Xco_2 value by the linear regression (Xco_{2est_i} , Eq. (1)) and the observed Xco_{2i} where i = observation number

$$Xco_{2R_i} = (\alpha - \Delta_i) - (\overline{Xco_2} - \alpha) \quad (\text{Eq. 3})$$

Where α = linear coefficient of the Xco_2 regression in a function of the observation; Δ_i = difference between the estimated value by regression and the observed value and $\overline{Xco_2}$ = mean value of all observed Xco_2 in the study region

After Xco_2 regionalization (Xco_{2R}), monthly averages were performed for Xco_2 , Xco_{2R} , SIF 757 nm, and Qg for the studied period. As regards precipitation and GPP, they were evaluated as monthly sums for the studied region. The averages were subject to analysis of variance (by Sdenecor's F-test at the 5 % probability level) to obtain the mean standard errors. In order to understand the variation of Xco_{2R} with the other variables, Spearman's and Pearson's correlation coefficients and subsequent simple linear regression analysis were calculated, both at the 5 % probability level for the t -test and F-test, respectively. Furthermore, the monthly time series was submitted to a Mann-Kendall test to evaluate whether there is some temporal trend in the Xco_{2R} . Finally, the annual averages of Xco_{2R} were compared to ECO_2 through linear regression.

2.4. Land use change and Xco_2

The Xco_{2R} was classified according to the land use in 2015, specifically in forest and farming areas, since they cover more than 90 % of the area. This was made by sampling the land use raster file to the observed coordinates of Xco_{2R} . To see the evolution of the Xco_{2R} and if the land use changes had some impact in this concentration, the 2015 land use classification was used (beginning of our study). For each land use (forest and farming) we applied a Mann-Kendall test to understand if there is a temporal trend in Xco_{2R} over an LU. It is important to notice that although Xco_{2R} theoretically has less atmospheric transport influence, due to the regionalization process, some uncertainty related to this process still remain in our analysis. Moreover, although CO_2 can be transported between both LU, at long-term trend we expect that this also can be reduced.

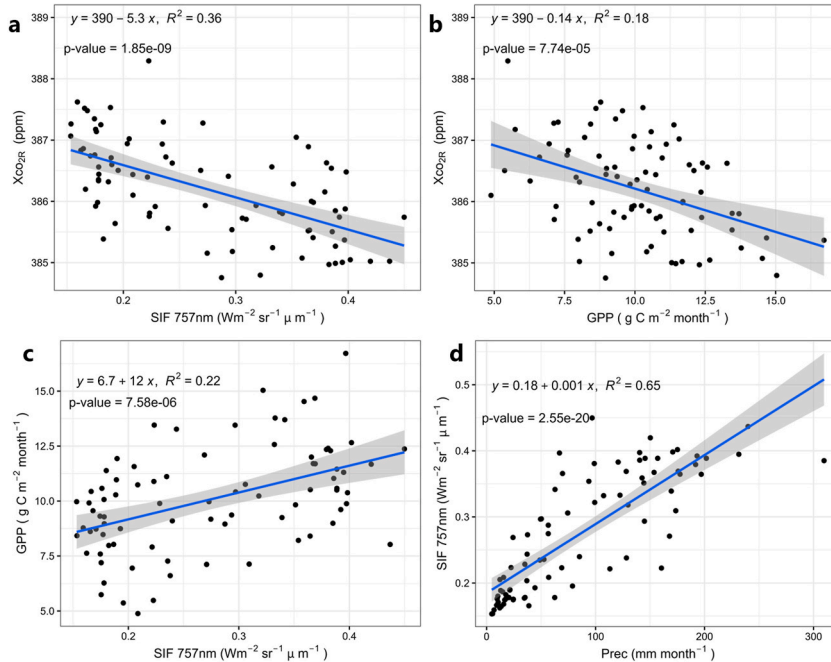


Fig. 4. Simple linear regressions between a) Xco_{2R} and SIF 757 nm, b) Xco_{2R} and GPP, c) GPP and SIF 757 nm, and d) SIF 757 nm and Prec. In which Xco_{2R} = regionalized column-averaged atmospheric CO_2 , SIF 757 nm = Solar-Induced chlorophyll Fluorescence at 757 nm, GPP = gross primary production, Prec = precipitation. Residuals related with these regressions are presented in SF1b to 1e.

3. Results

3.1. Temporal variability

The column-averaged CO₂ in the atmosphere (X_{CO_2}) of the Southeast region of Brazil, between 2015 and 2021, had a growth rate of 2.4 ppm year⁻¹ without detrending ($R^2 = 0.99$, $p < 0.0001$, Fig. 3a). After removing this global trend influence over time, we have a regionalized X_{CO_2} (X_{CO_2R} , Fig. 3a). Overall, the X_{CO_2R} seasonality had the highest means between May and October of each year and the lowest between November to April of each year (Fig. 3a). An important feature is that X_{CO_2R} presented a monotonic increase trend after 2017 which persisted even after removing the global tendency of around 0.01 ppm year⁻¹. This tendency is confirmed by a Pearson's correlation coefficient ($R = 0.42$, $p < 0.001$) and the Mann-Kendal test ($z = 1.99$, $p < 0.05$) and can be seen also by the regression line presented in Fig. 3a. Also, the X_{CO_2R} in 2016 is higher than in 2017, this could be associated with deforestation and afforestation trends in the period, further details will latter described in section 3.2.

The highest annual X_{CO_2R} averages were 392.26 ± 1.03 ppm (July 2020) and 393.40 ± 1.26 ppm (October 2021), mostly in winter and spring. Meanwhile, the minimum averages occurred in April 2015 (389.37 ± 1.7 ppm), the end of the summer season. Accordingly, the SIF 757 nm (Fig. 3b) presented its highest values in summer between December to April and a decrease between May and October, with the maximum value occurring in January 2021 (0.445 ± 0.24 Wm⁻² sr⁻¹ μm⁻¹) and minimum average in July 2018 (0.153 ± 0.11 Wm⁻² sr⁻¹ μm⁻¹). In general, the other variables behaved similarly to SIF. For example, the GPP (Fig. 3c) showed a maximum value of 16.72 kg C m⁻² in January 2019 and a minimum of 4.88 kg C m⁻² in October 2015. Similarly, the precipitation in this region, had a minimum of 4.8 mm in July 2018 and a maximum of 309.6 mm in January 2016 (Fig. 3e). Finally, the incident radiation (Fig. 3d) in the region had a maximum average in January 2019 (25.34 ± 3.95 MJ m⁻² day⁻¹) and a minimum average in June 2018 (13.81 ± 2.41 MJ m⁻² day⁻¹). This inverse behavior that X_{CO_2R} had with the other parameters is confirmed by the negative Spearman's correlation found in this study (SF2). Meanwhile, the other variables have a positive Spearman correlation between them, for example, SIF with precipitation (SF2).

The atmospheric CO₂ concentration decreases as SIF increases at a rate of decrease of 5.3 ppm for each unit of SIF, where this variable alone explains about 36 % of the monthly variability of X_{CO_2R} between 2015 and 2021 (Fig. 4a). This relationship is repeated with GPP, explaining about 18 % of the variability of X_{CO_2R} over time, where the rate of decrease with each unit of GPP is 0.14 ppm (Fig. 4b).

The SIF has a positive relation with GPP (Fig. 4c and SF2), with SIF alone explaining about 25 % of the variability of GPP over time, and where GPP grows at a rate of 0.12 kg C m⁻² month⁻¹ for each unit of SIF (Wm⁻² sr⁻¹ μm⁻¹). In addition, the SIF in the Southeast shows a growth rate of 0.001 for each mm of rainfall, with precipitation explaining about 65 % of the SIF variability in the region (Fig. 4d). Furthermore, similarly to the Spearman's correlation the X_{CO_2R} has a negative correlation with the other variables in the principal component analysis, thus an inverse direction (SF3).

3.2. X_{CO_2} and the land use

In the period of 2015 and 2021, land use changes (LUC) occurred over the whole region, but there is an accumulation of these changes especially in the North of Minas Gerais state (Fig. 5). Between 2015 and 2017, the afforestation over farming areas was around 7800 km² (~2.6 % of the total forest area in 2015). Conversely, the deforestation to farming expansion was about 5900 km² (~2 % of the total forest area in 2015). Meanwhile, between 2017 and 2021 there was a large deforestation to farming expansion and

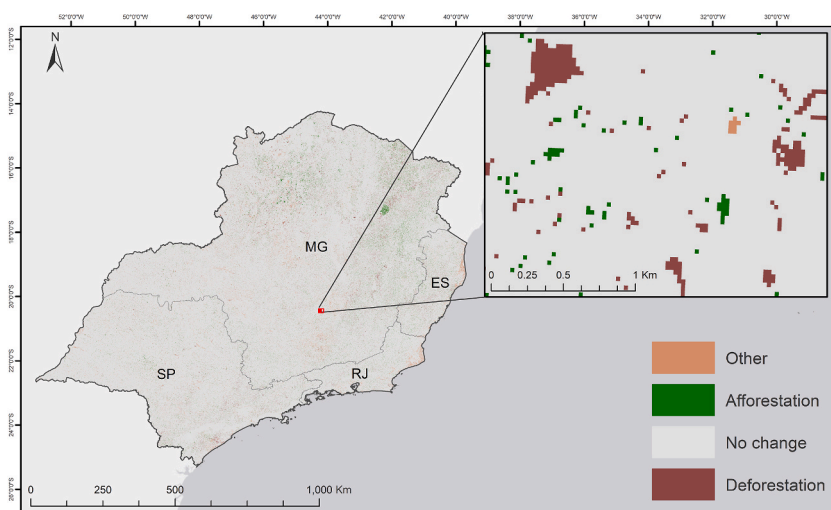


Fig. 5. Land Use Changes between 2015 and 2021 in Southeast; Afforestation - from farming (2015) to forest (2021), Deforestation - from forest (2015) to farming (2021). Inset amplification is an example of Land Use Change pattern.

afforestation with similar rates of around 14100 km² (~4 % of the total forest area in 2017). This means that between 2015 and 2017 the southeast had more afforestation than deforestation, and after that the deforestation was slightly higher than the afforestation, with a deforestation increase after 2017 (Table 2).

The Xco_{2R} between 2015 and 2016 was higher than compared with 2017 for both land uses (farming and forest) (Fig. 6a). Although deforestation rates between 2015 and 2017 due to farming were smaller than the posterior period (Table 2), is possible to notice that between these years, land use change emission estimates provided by SEEG showed a decrease, and after 2017 they started to rise again (Fig. 6b). Notably, the Xco_{2R} presented a growth tendency starting around 2017 (Figs. 3a and 6a). Spatial analysis over the studied region shows that the Xco_{2R} increase rate over forest areas was 30 % higher than over farming areas (Fig. 6a), confirmed by the Mann-Kendall test ($p < 0.05$). Moreover, there was a significant difference ($p < 0.05$) between the Xco_{2R} in the two different LU, with forest showing a higher average value.

The higher concentration and trend in forest land use could be ascribed by the deforestation due to farming expansion in the region. Since we classified the OCO-2 observation based on the land use of 2015, the deforestation after this period, especially after 2017, would potentially impact more the forest concentration. Also, since the farming areas are being afforested, the CO₂ removal capacity of forest formation could be keeping the increase rate lower. Although atmospheric transport could happen between both land uses and introduce some uncertainty, due to regionalization and the long-term trend analysis, this effect is diminished through time.

As we mentioned, the National inventory of CO₂ emissions provided by the SEEG (ECO₂, Fig. 6b) in Southeast Brazil presented a decrease of almost 15 M ton of CO₂ between 2015 and 2017, and after this period the emissions increased again more 15 M ton CO₂. This quantity left behind a major signal in the regional CO₂ concentration, which satellite long-term trend was able to capture. It is important to notice that the ECO₂ presented is only related to land use changes. Moreover, although deforestation is not the only source of these emissions, the main factor is the primary and secondary forest deforestation in the region (SF 4). Both types of emissions related to deforestation follow a similar pattern. Finally, the ECO₂ emission can explain around 50 % ($p < 0.1$) of the Xco_{2R} increase in the Southeast (Fig. 7), and with the release of 1×10^6 tons of ECO₂, this could imply an increase of 0.03 ppm in that region.

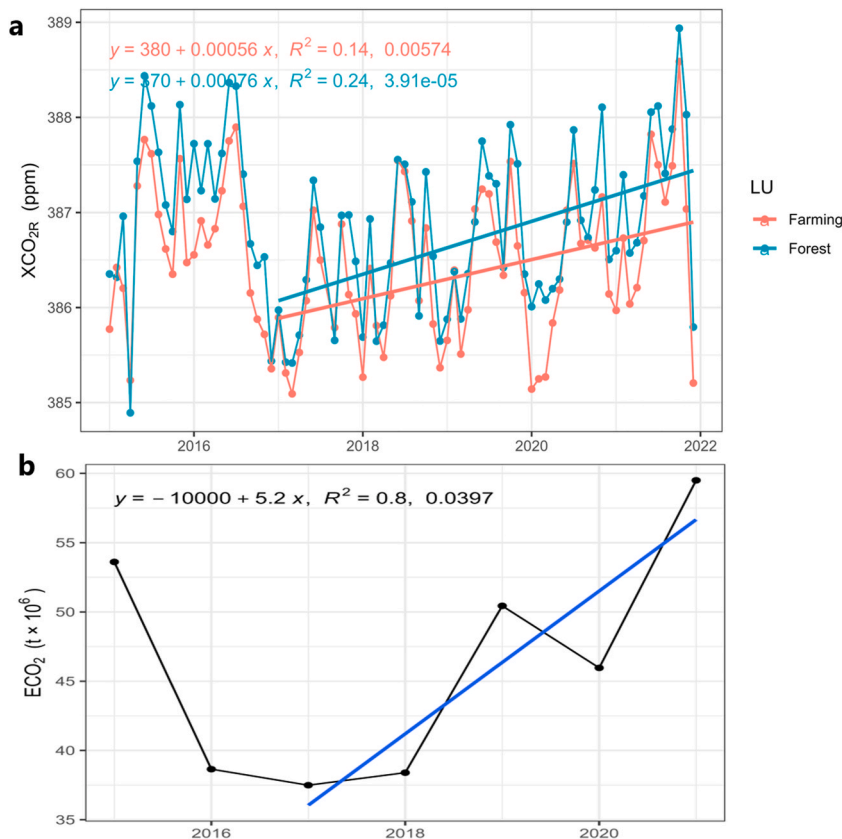


Fig. 6. Temporal variability of Xco_{2R} over different land uses (a) and of ECO₂ (b) during 2015 and 2021. The linear regressions show the trends after 2017 for both (Xco_{2R} and ECO₂). The p-values are inset after R². Residuals related with Xco_{2R} from farming and forest are presented in SF1f and 1g and the residuals of ECO₂ are presented in SF1h.

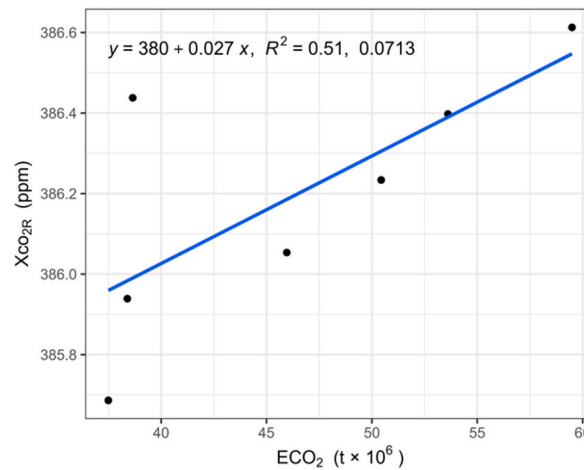


Fig. 7. Linear regression between annual Xco_{2R} and ECO₂ due to land use change for the study period (2015–2021). p-values are inset after R². Residuals are presented in SF 1i.

4. Discussion

4.1. Temporal variability

The atmospheric CO₂ concentration has been increasing over the decades and is one of the main causes of climate change (IPCC, 2022, 2021). This trend is also noticeable in the Brazilian Southeast (Xco₂ line, Fig. 3a), in agreement with what is being measured at ground stations (NOAA and AuthorAnonymous, 2024) and with satellite observations (Chen et al., 2022; Lopez et al., 2022). Furthermore, the Xco₂ (without the regionalization) does not correlate with any studied variable (SF2), and this is due to the global bias and the influence of atmospheric transport in the column-averaged atmospheric CO₂ undergoes (da Costa et al., 2022b; Wu et al., 2018). Given this, to study the natural and regional aspects of Xco₂ variability, it is necessary to detrend this curve (An et al., 2022; Hakkarainen et al., 2016; Rossi et al., 2022). Albright et al. (2022), in order to study the seasonal pattern over the Amazon, also detrend Xco₂ to exclude bias from anthropogenic sources and atmospheric transport, while da Costa et al. (2022b) made this for São Paulo State reporting that detrended Xco₂ did not have a significant correlation with wind velocity.

When we observe the Xco_{2R} we notice that it presents an inverse correlation with all the studied variables, especially the vegetative ones such as SIF 757 nm (SF2). This relationship has already been observed in other studies around the world (Albright et al., 2022; Parazoo et al., 2013; Santos et al., 2022; Shekhar et al., 2020). This inverse relationship is due to the photosynthetic process, which assimilates atmospheric carbon into biomass (Gu et al., 2019; Qiu et al., 2020; Taiz et al., 2015).

As SIF is a signal that originates from chlorophyll, it is considered one of the best approximations for the real state of plant photosynthesis (Mohammed et al., 2019; Porcar-Castell et al., 2014). We can observe the SIF link with photosynthesis through the GPP (Fig. 4c) since this variable can express the amount of carbon uptake by the plant and stored in the biomass. Several studies reported that SIF in a first approximation is linearly related to GPP (Damm et al., 2010; Frankenberg et al., 2014; Pacheco-Labrador et al., 2019; Rossini et al., 2010), as we observed here. However, this relation can be shaped by other mechanisms that we are not accounting for, such as the non-photochemical quenching and the leaf area (Köhler et al., 2018; Y. Liu et al., 2022; Martini et al., 2022; Pierrat et al., 2022; Rangel Pinagé et al., 2022; van der Tol et al., 2019). As photosynthetic activity of the vegetation present in the region increases, this process reduces the atmospheric CO₂ concentration (Chhabra and Gohel, 2019; Pierrat et al., 2022). This link explains the inverse relationship that we observe between Xco_{2R} with SIF 757 nm and GPP, as the CO₂ concentration decreases with the increase of both (Figs. 3 and 4). Additionally, the photosynthetic process is affected by climatic variations, such as precipitation and incident radiation (Barbosa et al., 2015; Liu et al., 2016). In their research, da Costa et al. (2023) studied the SIF variability in Brazilian biomes, observing that SIF is directly linked with precipitation and that the sensitivity to this phenomenon change according to the ecosystem. Similarly, we found a strong linear relationship between these two variables (Fig. 4d). Photosynthesis is also a water-dependent process, therefore when water availability increases, photosynthetic activity will also increase and in general this will be reflected in higher SIF values (Green et al., 2020; Jiao et al., 2019; Turner et al., 2021).

Regarding the Incident Irradiation (Qg), their influence on photosynthesis is direct, since part of the energy provided by Qg is used in the photosynthesis process. This is especially important for SIF and GPP since both are described by light models (Mohammed et al., 2019; Running et al., 2004). Thus, the amount of energy left behind by this biochemical process (SIF) and the amount of biomass produced (GPP), is directly related to the solar energy as observed here (SF2). This implies an inverse relation with the CO₂ concentration in the region (Fig. 3 and SF2), and similarly to our findings other studies also reported a similar pattern (da Costa et al., 2022b). However, this relation is not trivial, since not all of the solar energy is used by the plants, and even though the current methods to estimate the actual absorbed photosynthetically active radiation have improved, there is still high uncertainty in the remote sensing products regarding this energy (Pei et al., 2022).

Although climate plays an indirect relation in CO₂ concentration, as a driver of photosynthesis, the rainfall precipitation and the Incident Irradiation, also have other mechanism that interact with the atmospheric CO₂. For instance, the greenhouse effect is responsible for retaining part of the solar radiation and warming the planet. However, with the anthropic CO₂ emission, more carbon dioxide is added to the atmosphere, thus increasing global warming potential, and as a consequence, the global temperature average will increase (IPCC, 2022, 2021). Also, the CO₂ that is being added in atmosphere is modifying the optical depth. Higher optical depth induces that more light is scattered and less direct sunlight is reaching (Kerches Braghieri et al., 2020). Recently, Rodrigues et al. (2024) showed that the optical depth increase lead to a higher CO₂ net emissions in Amazon ecosystem. Despite CO₂ increases the optical depth, aerosols have higher impacts on it.

Moreover, the CO₂ concentration could affect the stomatal opening, and lead to changes in cloud formations and the water cycle (Berry et al., 2010; Driesen et al., 2020; Marchin et al., 2023; Sikma et al., 2019; Vilà-Guerau de Arellano et al., 2020). The stomatal closure and high vapor pressure deficit may contribute to higher carbon exchanges affecting latent and sensible heat (Vilà-Guerau de Arellano et al., 2020). The sensible heat reduction and latent heat increase reduces the turbulent kinetic energy negatively affecting cloud formations (Sikma et al., 2019). Recently, Luo et al. (2024) reported in a modelling study that deforestation can reduce cloud formation, and that besides this effect there is also an association between deforestation and rainfall reduction in tropical regions (Perugini et al., 2017). Furthermore, despite this cloud formation mechanism, rainfall can also be responsible to CO₂ decrease in atmosphere due to dilution (Ibáñez et al., 2022).

In general, the Xco_{2R} seasonality is closely related to the vegetation dynamics in the region, while in regions with intensive agricultural activity this cycle ends up being related to the agricultural year (da Costa et al., 2022a; Morais Filho et al., 2021). However, plant growth is driven by favorable climatic conditions, i.e., photosynthesis is a solar light and water-dependent process, and in the tropics the carbon uptake generally coincides with spring-summer and the maximum concentrations coincide with the fall and winter periods (Rossi et al., 2022). Morais Filho et al. (2021) also observed maximum Xco₂ in the fall-winter season and predominantly over agricultural regions, recalling that this season is characterized by an intense land management activity in Brazil. This is an indication that during the fall-winter season, the lack of correct soil management, such as minimal soil disturbance, can substantially contribute to the increase in the atmospheric CO₂ levels beyond the natural cycle (Panosso et al., 2009; Perillo et al., 2022).

4.2. Land use change and Xco₂

It is a major challenge to establish and specify links and mechanisms between land use conversion and CO₂ emission within the agroecosystem complexity. When it comes to land use dynamics in Brazilian Southeast, that was historically shaped by farming over native vegetation (de Mendonça et al., 2022; Ribeiro et al., 2009; Souza et al., 2020; Strassburg et al., 2017; Trigueiro et al., 2020); carbon fluxes reflect the multifunctional context of water-energy-food-land-climate nexus (Shi et al., 2023). Where deforestation-risk farming in tropical countries plays a key role in global conservation governance (Garrett et al., 2022). Therefore, the methods and results reported here also present subnational trends in CO₂ emissions related to land use change to support further mitigation strategies and zero-deforestation targets in agricultural supply chains.

As highlighted, we observed that around 14 thousand km² of farming areas replaced forest covers in a deforestation movement between 2021 and 2017 (Table 2). According to land use monitoring systems, PRODES Cerrado (INPE, 2019) and MapBiomass (RAD-MapBiomass, 2021), both biomes, Cerrado and Atlantic Forest, are facing an increase in deforestation rates, which intensified between 2019 and 2021 – ranging from 25 % to 66 %. The Xco_{2R} increase rates show important trends regarding the variability of CO₂ concentrations and deforestation in the region. As reported in recent land cover–atmosphere modeling, CO₂ emission in agroecosystem is strongly associated with agricultural activities and forest cover loss in Brazil (Crivelari-Costa et al., 2023; Garofalo et al., 2022).

Between 2015 and 2017, the afforestation was higher compared to deforestation (Table 2), and this less deforestation led to a decrease in the ECO₂ (Fig. 6b), contributing to the slightly lower Xco_{2R} levels between 2016 and 2017 (Figs. 3a and 6a). However, after 2017 there was an increase in ECO₂ related to land use change (Fig. 6b), especially due to deforestation for farming expansion, explaining the trend in Xco_{2R} and the higher rate over forest (Figs. 3a and 6a). According to the Brazilian System for Greenhouse Gas Emissions and Removals Estimates (SEEG, De Azevedo et al., 2018), the last data release pointed out that there was an increase in the CO₂ equivalent emissions associated with land use change for the same period (2017 and 2021). These emissions are especially due to deforestation for farming expansion. In 2020, Brazil's land use change and forestry sector corresponded to an estimated 3.3 ± 2.6 GtCO₂ yr⁻¹ emissions, around 66 % of the country's CO₂ emissions (SEEG, 2021) and 9 % of global CO₂ emissions (Friedlingstein et al., 2022).

Through the absorption, storage, and release of various carbon amounts, land use plays a key role in the carbon cycle and consequently in atmospheric CO₂ amounts (Siabi et al., 2019). Land use change and the forestry sector are important climate change drivers, different land cover exerts important impacts on CO₂ emission which substantially increases in farmland over forest areas, thereby altering the spatiotemporal variations and even regional and global warming (Friedlingstein et al., 2001; J. Liu et al., 2021).

Table 2
Land use changes in km² due to deforestation for farming expansion and afforestation over farming areas between 2015 and 2021.

Period	Afforestation	Deforestation
2017-2015	7835.872	5936.135
2021-2017	14118.26	14195.48

In addition, forest fragmentation makes native vegetation more vulnerable to disturbances and damage to ecosystem functions, especially in buffer zones with less advanced successional stages (Junior et al., 2018; Parras et al., 2020). Furthermore, interannual and seasonal trends in the vegetation of the Atlantic Forest and Cerrado Biomes are observed concerning the influence of climatic variables (R. G. da Silva et al., 2021). Likewise, deforestation can increase the risk of fire incidence and lead to additional CO₂ emissions (Alencar et al., 2022; Aragão et al., 2018; Gatti et al., 2021). According to Guedes et al. (2020), the risk of exposure to fire is greater in stands with canopy coverage of less than 20 %, especially within 180 m of the Atlantic Forest patch edges. In Cerrado vegetation, aspects of fire behavior and fire-associated carbon emissions differ between vegetation types, but compared to forest typologies, natural savannas, and larger pastures experienced higher rates of fire transmission and intensity, in addition, agricultural frontiers increase combustion sources.

Despite our work focuses in the land use change – CO₂ patterns nexus, fire outbreaks are also pivotal for Brazilian CO₂ emission profile. Between 1999 and 2018, the cumulated contribution from fires to atmosphere was more than 8 Gton of carbon and was projected that between 2019 and 2030 the cumulated emission would be around 5 Gton CO₂ (C. A. da Silva et al., 2020). Gomes et al. (2024) recently showed 1.5 Gton of the emissions between 1985 and 2020, only in Cerrado. Accordingly, with SEEG (2021) Minas Gerais State was the one with higher CO₂ emissions related with fire ($55.7 \times 10^3 \pm 18.2 \times 10^3$ ton CO₂ e), followed by São Paulo ($14.8 \times 10^3 \pm 4.8 \times 10^3$ ton CO₂ e) (Supplementary Table 1). However, for São Paulo State, the major CO₂ source is related with energy sector ($\sim 89 \pm 4.5$ M ton CO₂).

Although fire is an important CO₂ source, compared with deforestation emissions (Fig. 6b), the signal left behind by this source would be suppressed by the deforestation signal, especially considering the monthly/annual averages analyzed here. Moreover, the emissions related with energy sector, that are extremely important for São Paulo, are almost constant during the analyzed period and, this emission category is more of a point source. Due to these characteristics, the detrending and space-time aggregation would reduce the influence of this sector in the signal here analyzed.

The forest cover increased by around 21 thousand km² (between 2015 and 2021) over farming uses (Fig. 5 and Table 2). This could be ascribed to ongoing efforts to restore and recover native vegetation, either through federal instruments and policies or through non-governmental and civil society initiatives for environmental conservation (Brancalion et al., 2016; Colman et al., 2022; Crouzeilles et al., 2019; de Mendonça et al., 2022; Melo et al., 2013). Reforestation and natural revegetation play a very active role in carbon (C) sinks in degraded ecosystems. However, the responses of C sequestration and stabilization, either in biomass or soil organic matter, will only be perceived in the long term (Yang et al., 2023).

Although the area of deforestation by farming and reforestation changed more or less the same in total area (Table 2), the carbon emission and removal dynamics do not show the same balance. Even with the benefits from degraded pasture recovery or conversion of agricultural land to forest restoration, tree carbon absorption capacity ranges from 2 to up of 12 Mg ha⁻¹ year⁻¹ of C until reaching the climax (Ahmed et al., 2024; Bai and Ding, 2024; Pérez-Cruzado et al., 2012). The time needed for a secondary forest to reach a stable climax stage, pushing to a net zero by removing and stocking carbon, can take about 25 years in tropical forests (Sahu et al., 2022). We reinforce that we detected this change between 2015 and 2021, but we did not address the time that the change occurred, which could imply an overestimation of the forest age in our study.

The Xco_{2R} has an increasing trend in both land uses studied here (farming and forest) due to the land use changes that occurred after 2017 (Fig. 6). The replacement of forest and natural covers by farming areas does not necessarily represent a photosynthesis change, as we can notice that the SIF and GPP didn't have a decreasing trend (Fig. 3). However, this could represent a CO₂ emission from the soil. The LUC is an intensive activity that changes and controls soil structure and quality, especially deforestation (Leul et al., 2023; Mukhopadhyay et al., 2023); besides modifying the stoichiometry of soil nutrients and carbon and greenhouse gas emissions (X. Liu et al., 2021).

5. Conclusion

The Xco₂ is influenced by global bias and atmospheric transport, meaning that to access the impact of regionalized studies they must be detrended (Xco_{2R}). Xco_{2R} generally has greater mean values during autumn-winter and lower during spring-summer, with an inverse relation with vegetation and climate parameters. This is because the more favorable climate conditions in spring-summer enhance the photosynthesis rate and lead to a decrease in the atmospheric CO₂ in the region, thus acting as a carbon sink in this season in Southeast Brazil.

We found that, after 2017, the Xco_{2R} presented a significant increase even detrending Xco₂ global bias. This was also observed in forest and farming areas, and in the first land use (forest) this trend was stronger possibly because of a deforestation increase after 2017. We found that Xco_{2R} is directly related to CO₂ equivalent emission from land use changes reported in national GHG inventories due to land use change, especially deforestation. Also, forest reforestation in the period was not sufficient to recover the atmospheric CO₂ accumulation trend especially due to deforestation from 2017 to 2022. Therefore, more efforts on the knowledge of seasonality and forest phytophysiology could improve net-zero strategies and help atmospheric CO₂ decrease in the tropics in the near future.

Moreover, our study is limited due to the disregarding of atmospheric transport and other CO₂ sources, such as fire emissions and fossil fuel combustion in large cities. Despite the limitations, we diminished the atmospheric transport influence through the detrending process, as our goal was to study local influence in the CO₂ seasonality. Also, land use change is underestimated in our work, since we are only considering the LUC related to farming and forest cover types.

CRediT authorship contribution statement

Luis Miguel da Costa: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Gislaine Costa de Mendonça:** Data curation, Formal analysis, Writing – review & editing. **Gustavo André de Araújo Santos:** Data curation, Formal analysis, Supervision, Writing – review & editing. **Fernando Antônio Leal Pacheco:** Data curation, Methodology, Writing – review & editing. **Glauco de Souza Rolim:** Data curation, Methodology, Writing – review & editing. **Alan Rodrigo Panosso:** Data curation, Supervision, Writing – review & editing. **Newton La Scala Jr.:** Conceptualization, Data curation, Methodology, Supervision, Writing – review & editing.

Code and data availability

The codes used and the data pre-processed are available at github: https://github.com/lm-costa/xco2_drivers_2023; the full OCO-2 data is available at: https://disc.gsfc.nasa.gov/datasets/OCO2_L2_Lite_FP_10r/summary; the data retrieved from AppEEARS can be requested at: <https://appeears.earthdatacloud.nasa.gov/>; the NASA-POWER data can be requested at: <https://power.larc.nasa.gov/>; The land use classification are available at: <https://mapbiomas.org/>; The CO₂ equivalent emission data are available: <https://seeg.eco.br/>

Ethical statement

All ethical practices have been followed in relation to the development, writing, and publication of the article. In so doing, the authors confirm that they have followed the regulations of our institutions concerning intellectual property.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rsase.2025.101614>.

Data availability

All codes and data used and their respective access links are available through the main text

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